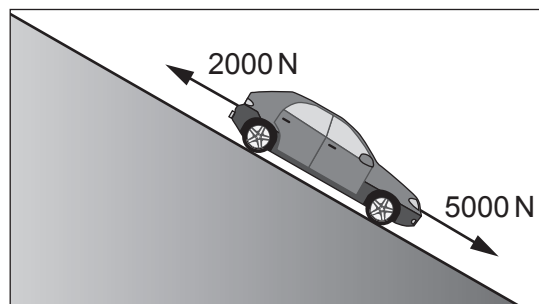


9. The diagram shows a car rolling down a slope. Two of the forces acting on the car are labelled.



- (a) The car has a **weight** of 10 000 N. Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the car.  
(Gravitational field strength,  $g$ , = 10 N/kg)

[2]

mass = ..... kg

- (b) Use the information in the diagram to answer the following questions.

- (i) Calculate the resultant force acting down the slope.

[2]

resultant force = ..... N

- (ii) Use your answers from parts (a) and (b)(i) and the equation:

$$\text{acceleration} = \frac{\text{resultant force}}{\text{mass}}$$

to calculate the acceleration of the car at the instant shown and state the unit. [3]

acceleration = .....

unit = .....



- (iii) I. Explain how the resultant force on the car changes as it speeds up. [2]

.....

.....

.....

- II. State how this change in resultant force affects the acceleration of the car. [1]

.....

- (c) At the bottom of the slope the car continues horizontally at a constant speed of 12 m/s with a kinetic energy of 72 000 J.

- (i) State **one** reason why the potential energy at the top of the hill must have been greater than 72 000 J. [1]

.....

- (ii) At the bottom of the hill a braking force is applied which stops the car over a distance of 15 m.

Use the equation:

$$\text{force} = \frac{\text{work done}}{\text{distance}}$$

- to calculate the braking force. [2]

braking force = ..... N

**END OF PAPER**

